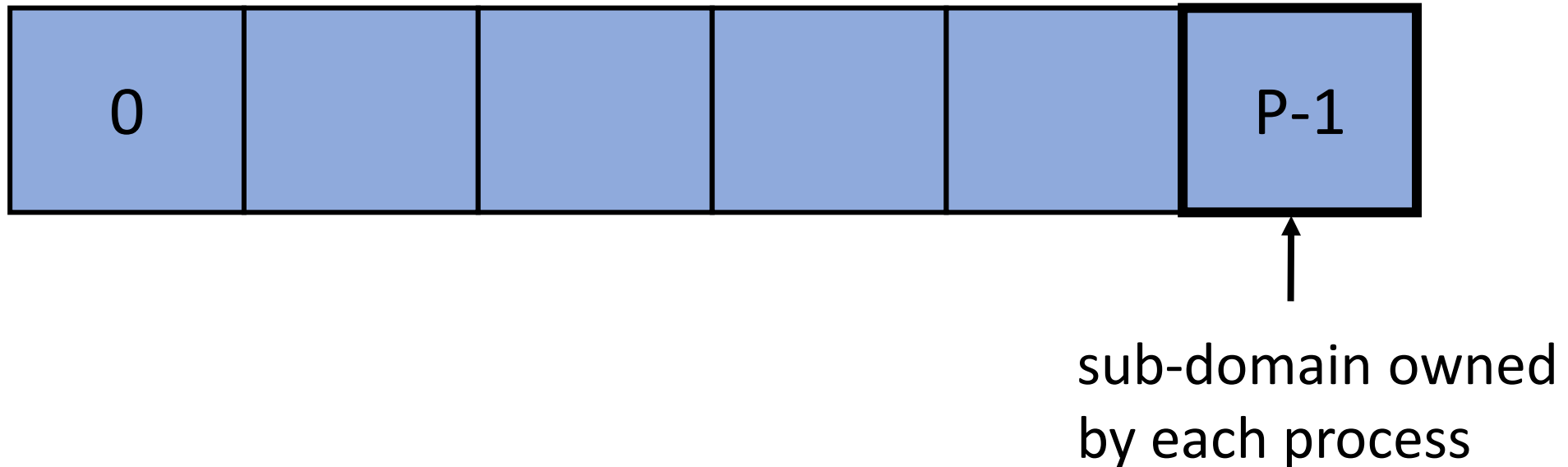


Communication

Lecture 8

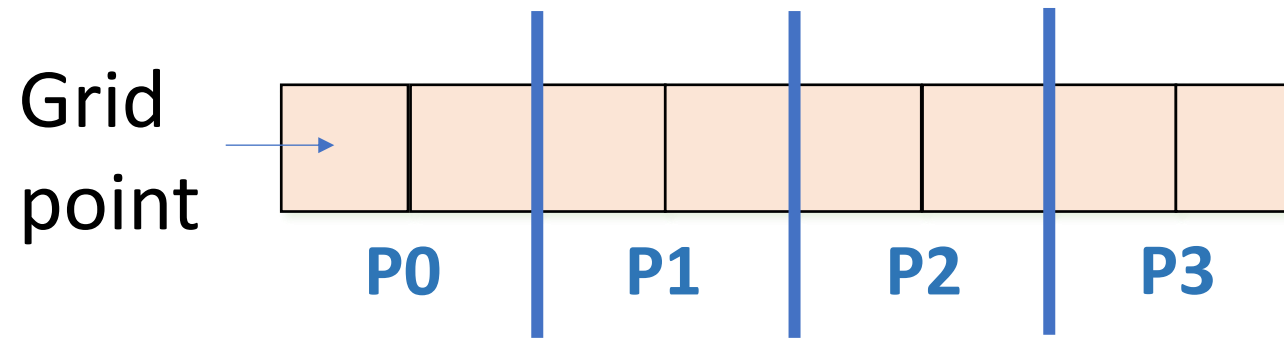
February 2, 2026

1D Domain and 1D Domain Decomposition



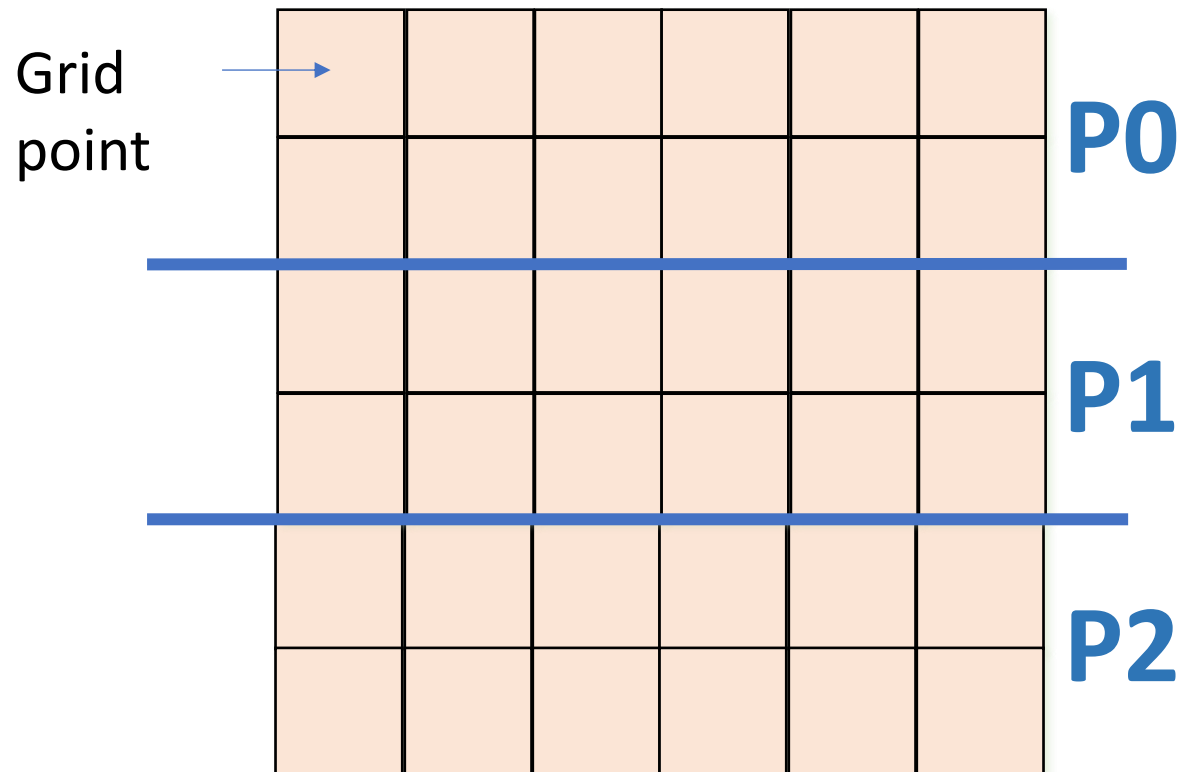
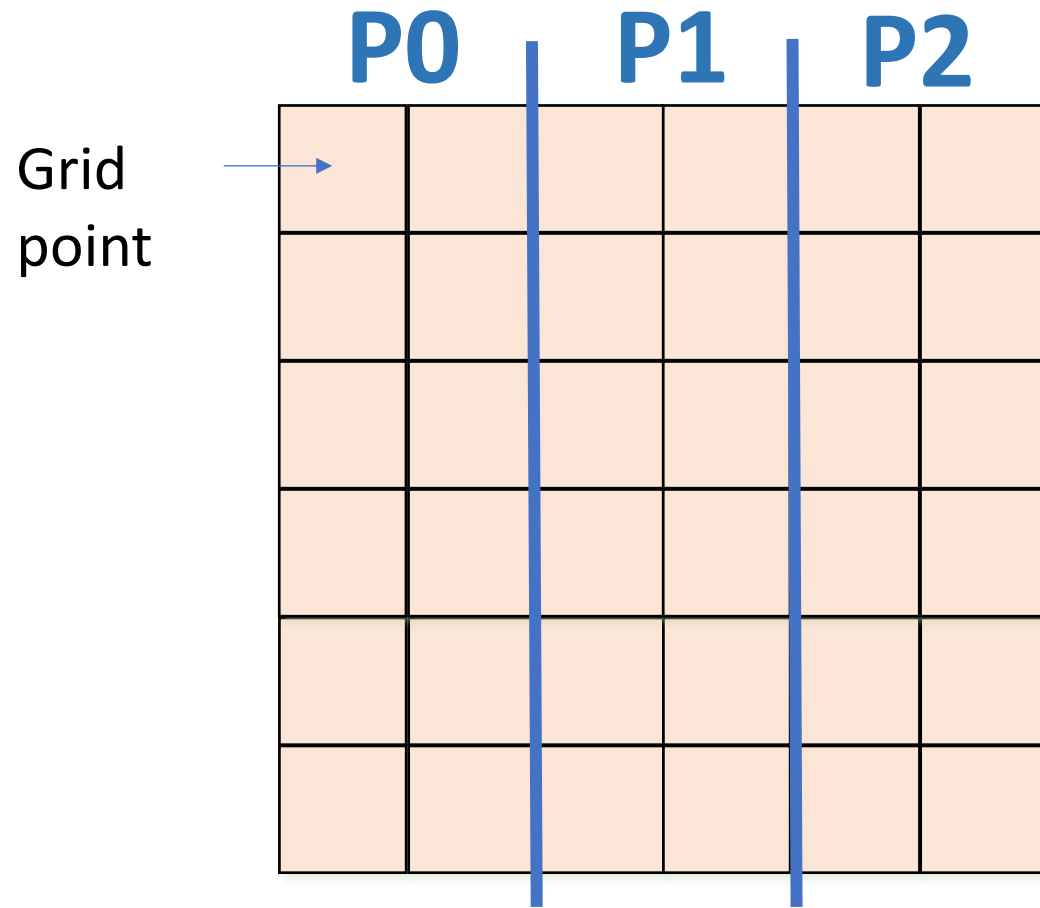
- 1D domain equally sub-divided into P processes
- One of the **communication patterns** may be NN

1D Domain Decomposition

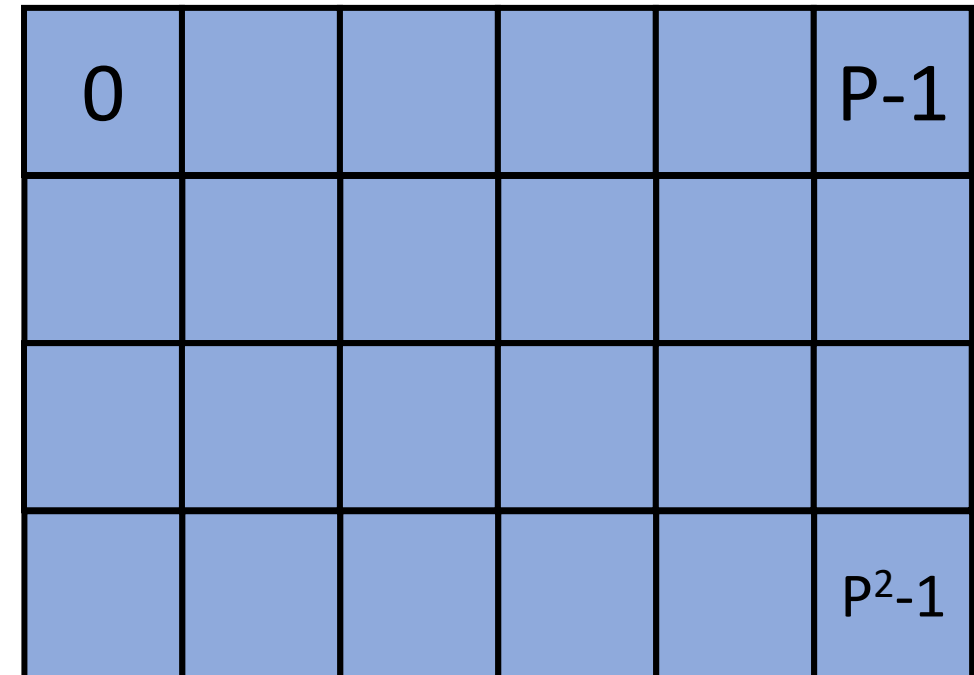
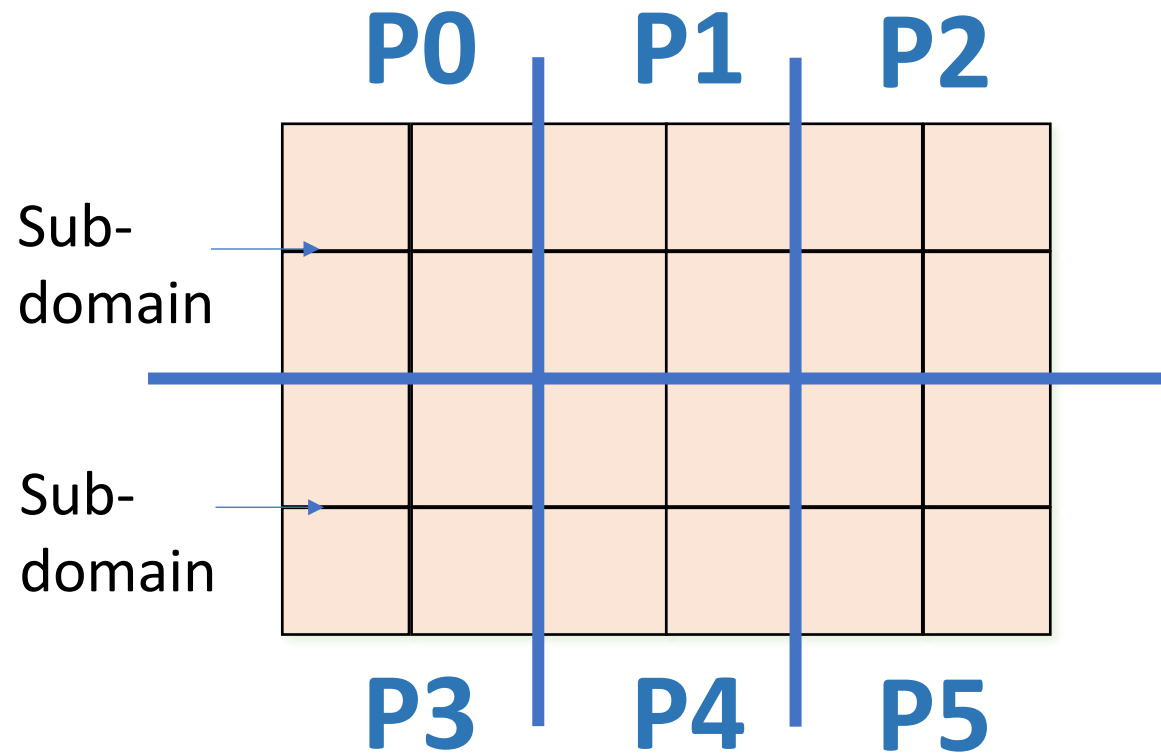


1D Process Decomposition
1D Domain (e.g. 1D array)

1D Decomposition of 2D Domain (e.g. 2D array)

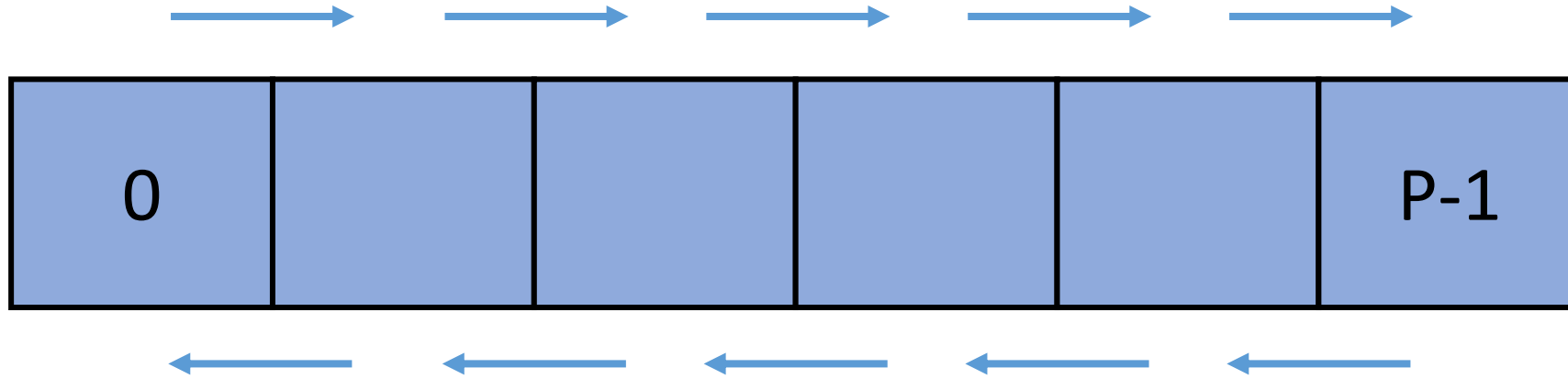


2D Domain Decomposition of a 2D Domain



XY process decomposition

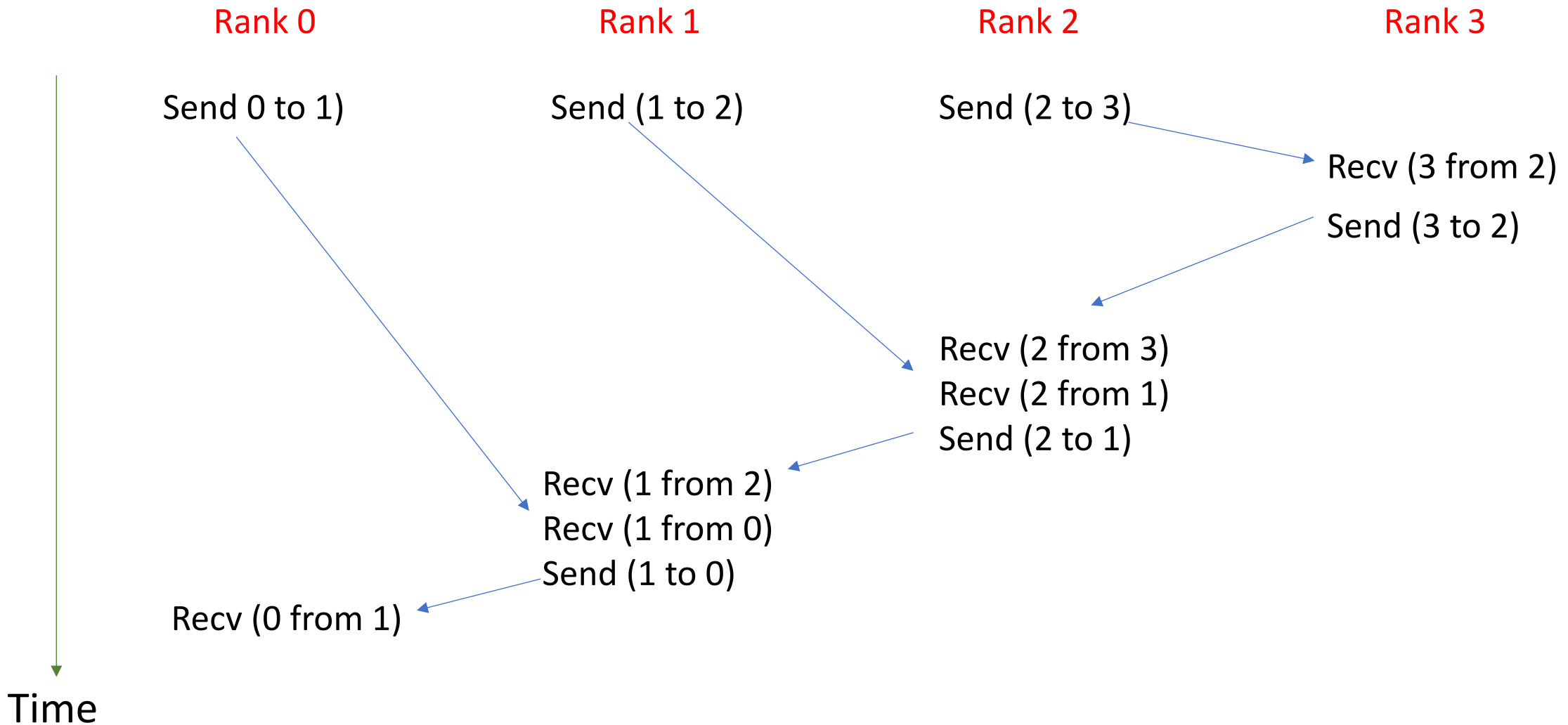
Nearest Neighbor (NN) 2-way 1D Exchange Recap



Option 1: Schedule $R \rightarrow$ sends, then $L \leftarrow$ sends

```
if (myrank < P-1)
{
    MPI_Send (data, myArraySize, MPI_DOUBLE, myrank+1, myrank+1, MPI_COMM_WORLD);
    MPI_Recv (recvbuf, myArraySize, MPI_DOUBLE, myrank+1, myrank, MPI_COMM_WORLD, &status);
}
if (myrank > 0)
{
    MPI_Recv (recvbuf, myArraySize, MPI_DOUBLE, myrank-1, myrank, MPI_COMM_WORLD, &status);
    MPI_Send (data, myArraySize, MPI_DOUBLE, myrank-1, myrank-1, MPI_COMM_WORLD);
}
```

Example: 4 processes

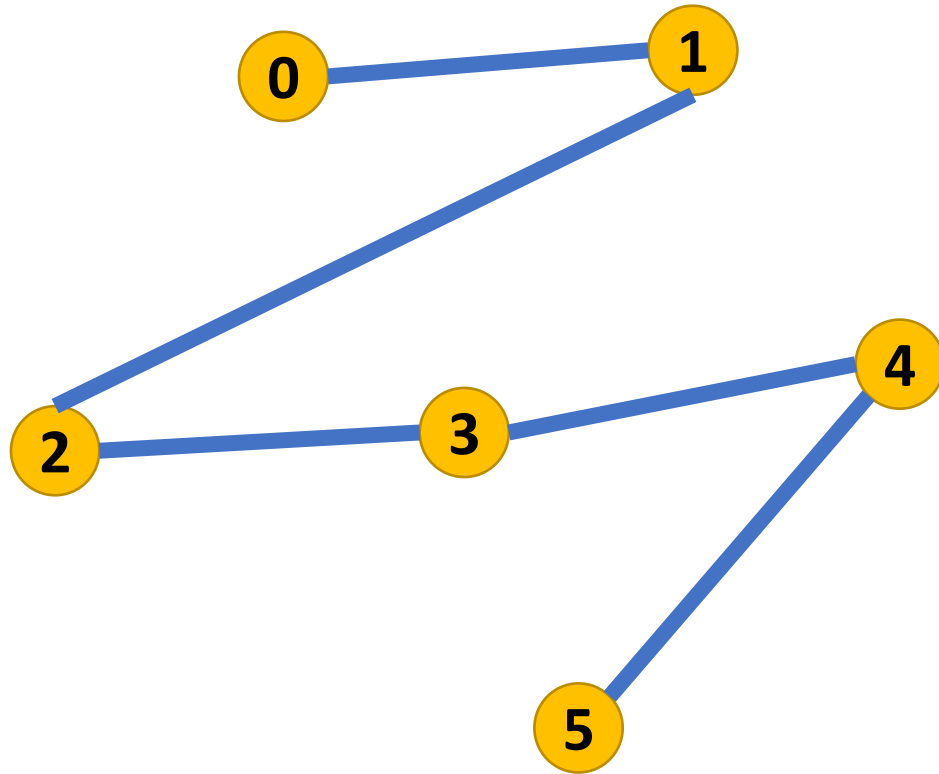


Option 2: Odd-even/Even-odd Scheduling

```
if (myrank % 2 == 0 && myrank < P-1) {  
    MPI_Send (... , myrank+1, myrank+1, ...);  
    MPI_Recv (... , myrank+1, myrank, ...);  
}  
else if (myrank % 2 != 0 && myrank > 0) {  
    MPI_Recv (... , myrank-1, myrank, ...);  
    MPI_Send (... , myrank-1, myrank-1, ...);  
}
```

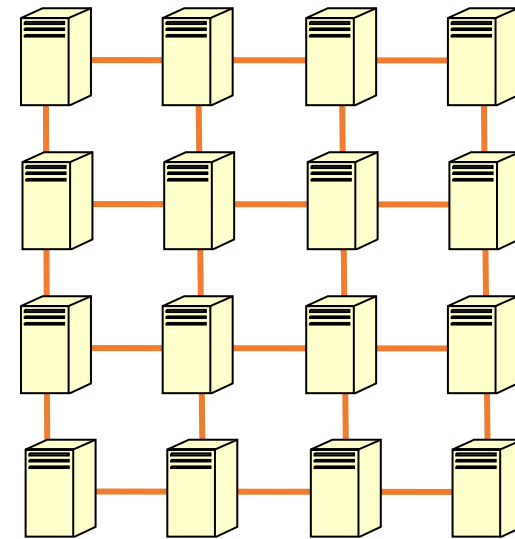
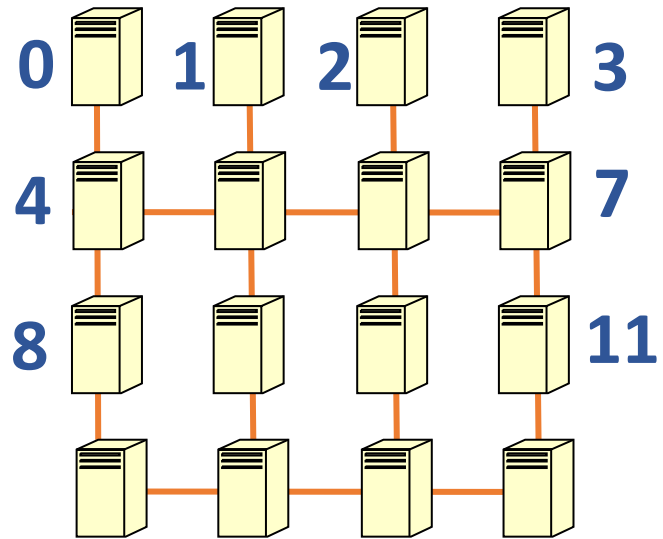
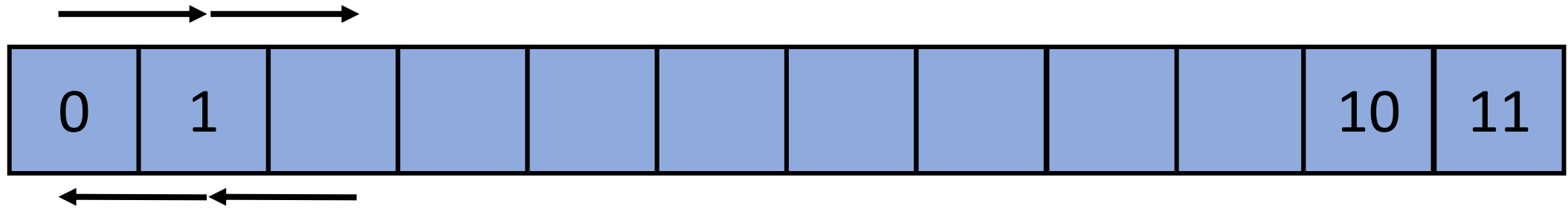
```
if (myrank % 2 != 0 && myrank < P-1) {  
    MPI_Send (... , myrank+1, myrank+1, ...);  
    MPI_Recv (... , myrank+1, myrank, ...);  
}  
else if (myrank % 2 == 0 && myrank > 0) {  
    MPI_Recv (... , myrank-1, myrank, ...);  
    MPI_Send (... , myrank-1, myrank-1, ...);  
}
```

Communication Graph/Matrix for 2-way NN

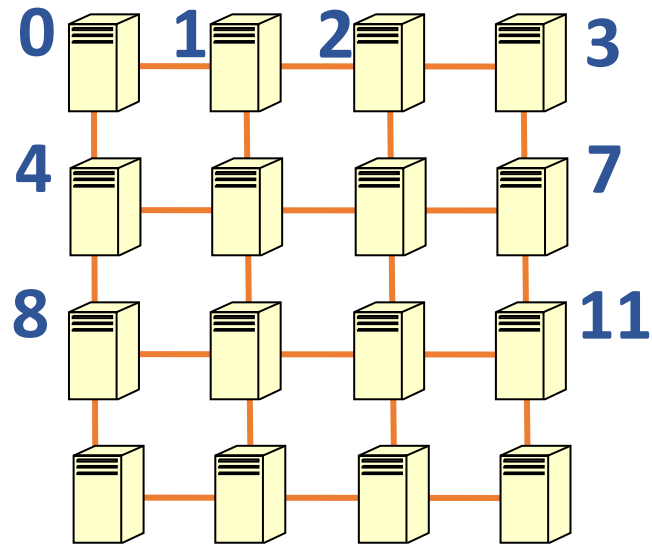
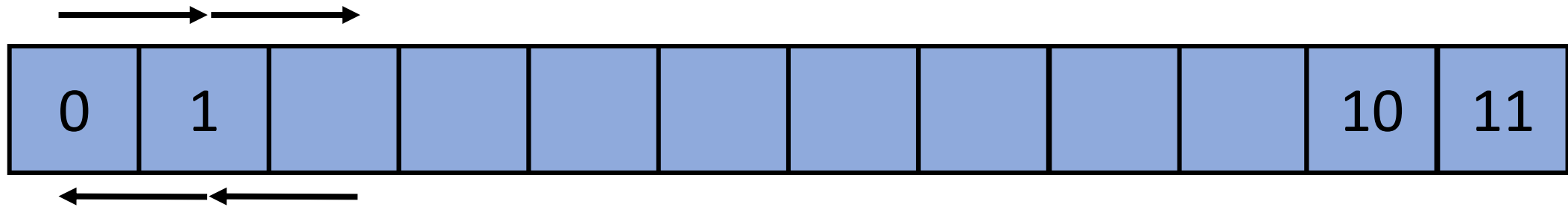


	0	1	2	3	4	5
0		512				
1	512		512			
2		512		512		
3			512		512	
4				512		512
5					512	

Process Mapping/Allocation

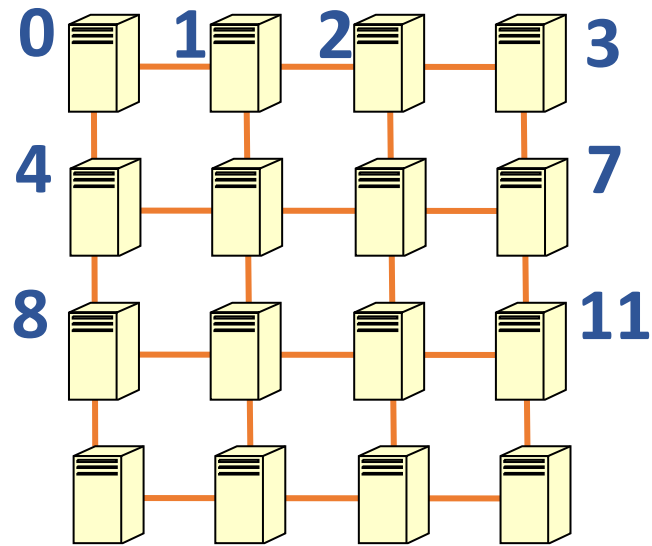
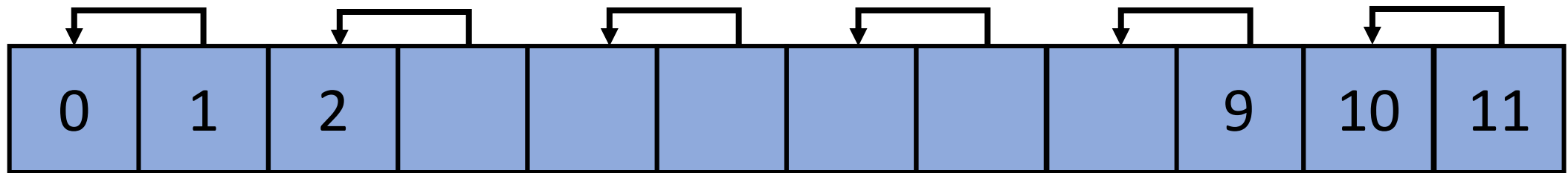


Process Mapping/Allocation



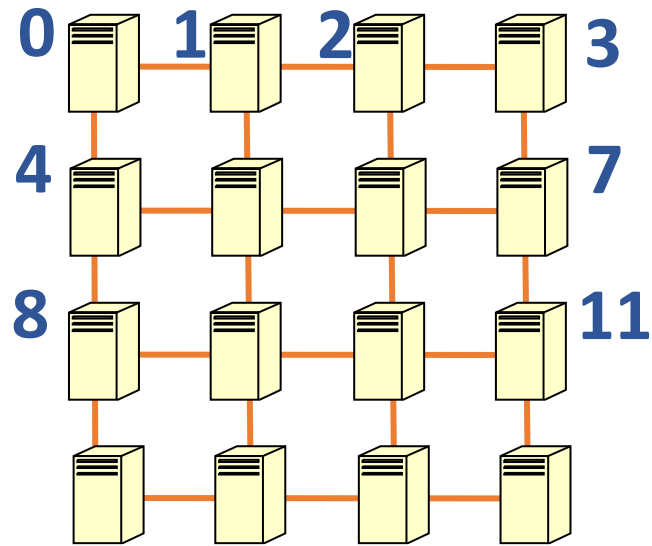
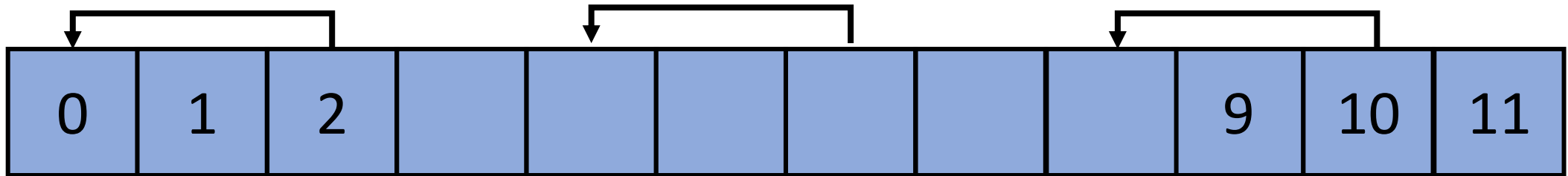
- This network may result in less average number of hops for the above communication pattern
- This network has higher cost than the other network (had fewer links)

Process Mapping/Allocation



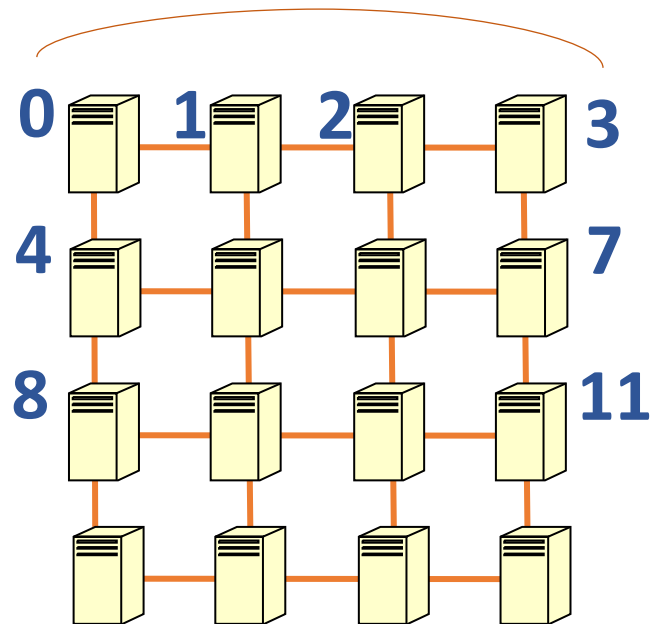
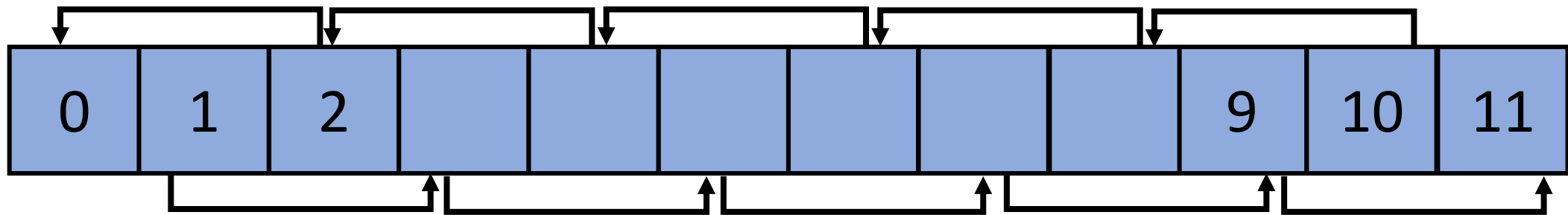
- Maximum number of hops?
- Average number of hops?

Process Mapping/Allocation



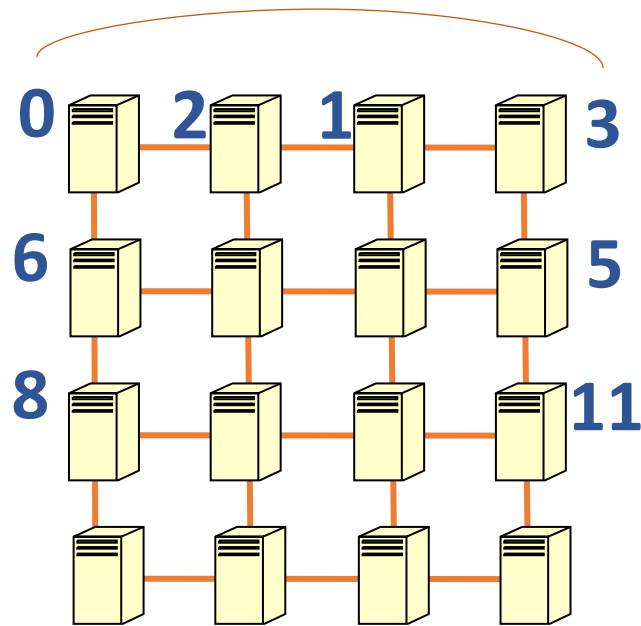
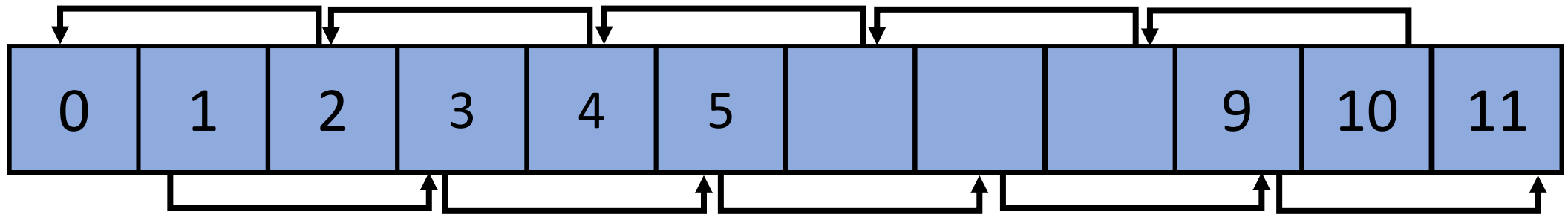
- Maximum number of hops?
- Average number of hops?

Process Mapping/Allocation

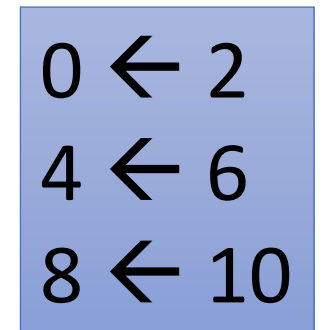
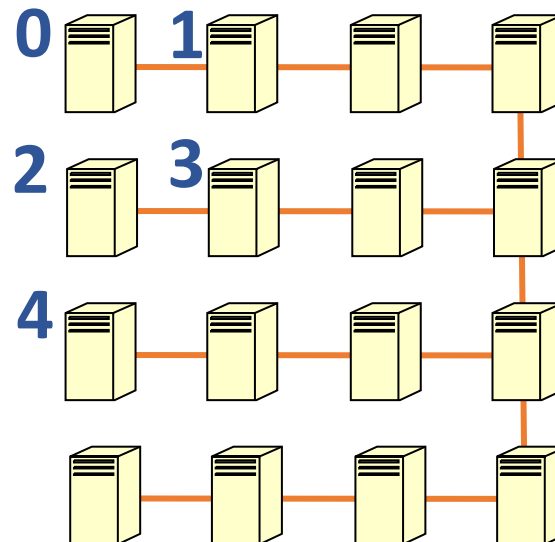
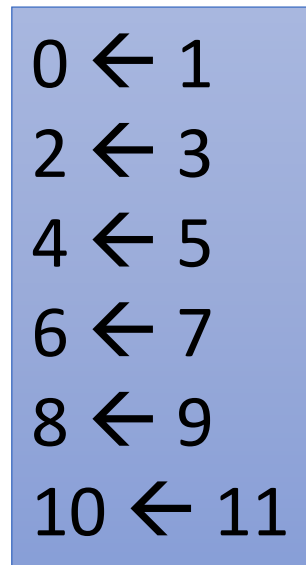
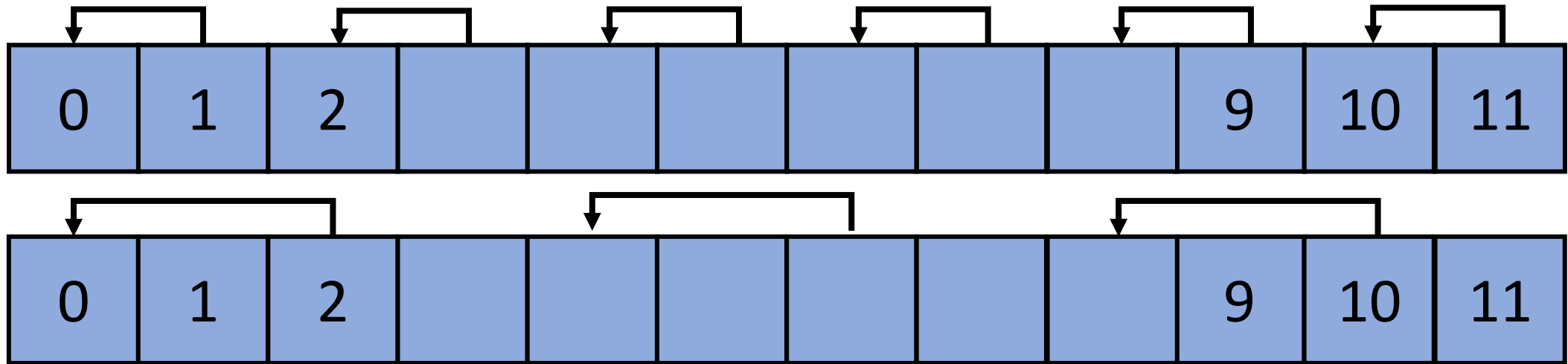


- (1) Maximum number of hops?
- (2) Maximum number of hops in modified network?
- Suggest a process mapping where maximum number of hops = 1

Topology-aware Process Mapping



Virtual Topology to Physical Topology



Interconnects

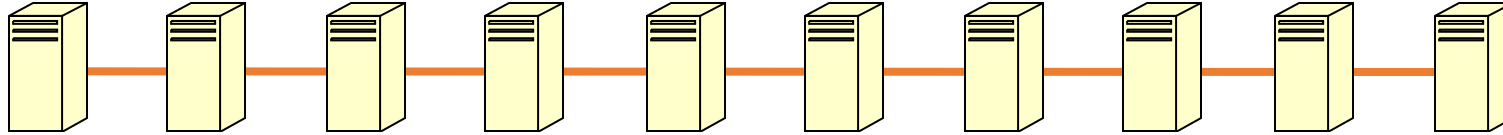
Attributes of Interconnects

- Topology
- Diameter
- Cost
- Anything else?

Attributes/Characteristics of Interconnects

- Topology
- Diameter
- Cost
- Bisection width

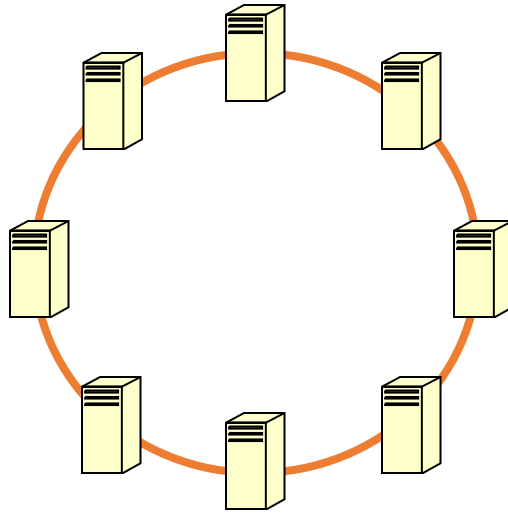
Linear Array



Total #nodes = p

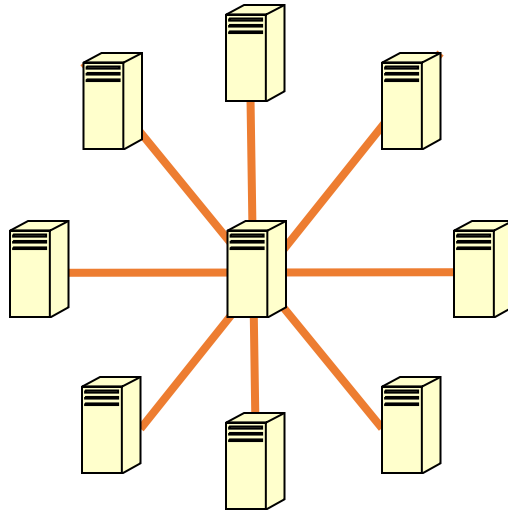
- Diameter $p-1$
- Bisection width 1
- Cost $p-1$

Ring Interconnect



- Diameter $p/2$
- Bisection width 2
- Cost p

Star Interconnect

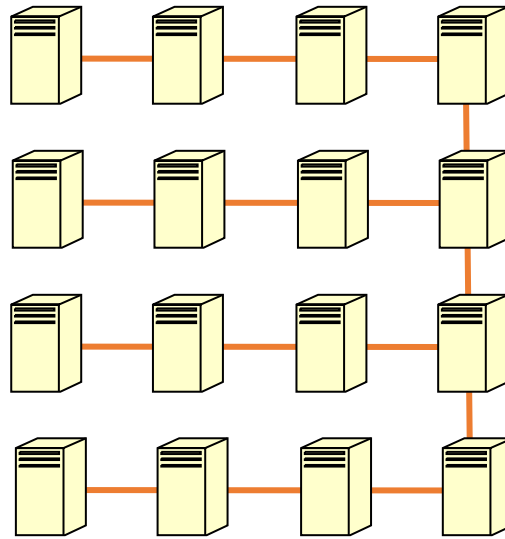


- Diameter 2
- Cost $p-1$

Drawback?

Suited for?

Hypothetical Network



Hands-on

- P2rutor
 - https://docs.google.com/document/d/160H4OQwJHW3q_AMndHLAwzXXfD0n_cmo/edit?usp=sharing&oid=106578722130257784179&rtpof=true&sd=true
 - <http://172.27.21.233:82/>
- PARAM Rudra
 - https://docs.google.com/document/d/1-bhmmwO23PGDEFloQMRegeqAGgJyhTiyYQqb2dlOe_w/edit?usp=sharing
 - ssh username@paramrudra.cdacdelhi.in -p 4422 -o ServerAliveCountMax=15 -o ServerAliveInterval=15
 - Use MobaXTerm on Windows